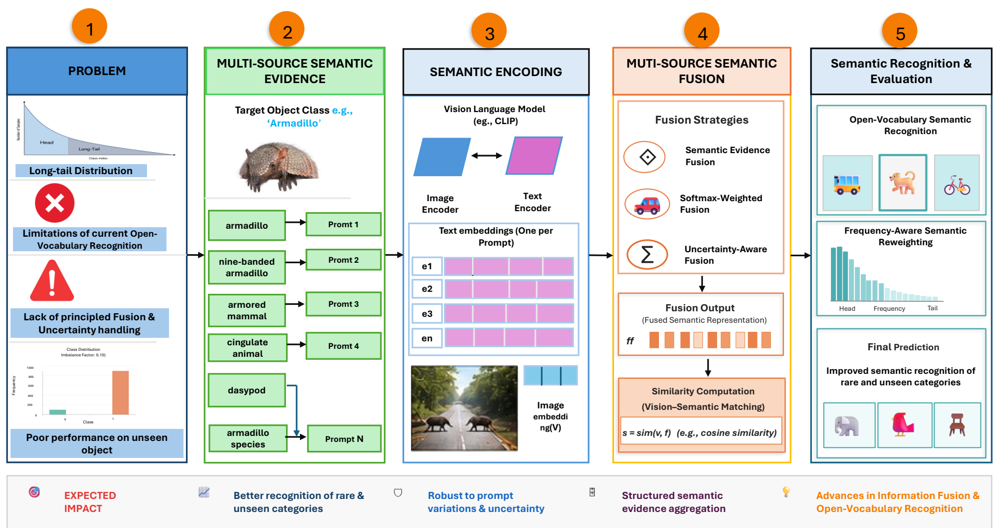
Graphical Abstract

When Does Semantic Prompt Fusion Help? A Critical Multi-Source Evidence Analysis for Long-Tail Open-Vocabulary Object Detection

Haripriya Chalicheemala, Raja Hashim Ali, Iftikhar Ahmed



Highlights

When Does Semantic Prompt Fusion Help? A Critical Multi-Source Evidence Analysis for Long-Tail Open-Vocabulary Object Detection

Haripriya Chalicheemala, Raja Hashim Ali, Iftikhar Ahmed

- Treating Language as Evidence: A New Paradigm for Rare Object Detection
- Adaptive Fusion: Making AI Smarter About What It Doesn't Know

When Does Semantic Prompt Fusion Help? A Critical Multi-Source Evidence Analysis for Long-Tail Open-Vocabulary Object Detection

Haripriya Chalicheemala[a], Raja Hashim Ali[a], Iftikhar Ahmed[a]

[a]*Department of Business, University of Europe for Applied Sciences, Think Campus, Konrad-Zuse-Ring 11, Potsdam, 14469, Brandenburg, Germany*

Abstract

Object detection systems are widely used in autonomous driving, robotics, surveillance, industrial automation, as well as in medical imaging. In real-world datasets, however, the object categories are unevenly distributed, as some classes contain thousands of training examples while others contain only a few samples. This long-tail distribution leads to a significant performance loss for rare and previously unknown object categories especially when conventional deep learning models are trained on datasets dominated by frequently occurring classes. Recent vision-language models such as CLIP have improved open-vocabulary object detection by learning semantic relationships between textual descriptions and visual content. Existing methods, however, rely mainly on individual prompts or simple averaging strategies, which do not fully utilize the semantic diversity available through synonyms, taxonomic relationships, and descriptive paraphrases. This study presents a framework for semantic fusion from multiple sources, treating semantic prompts as independent sources of evidence and combining them using various fusion strategies. The proposed framework integrates Mean Fusion, Max Fusion, Weighted Fusion, and Adaptive Fusion with uncertainty-aware semantic modelling. The semantic prompts are generated using WordNet-based synonym extraction along with LLM-based paraphrasing techniques. To improve robustness for rare categories, entropy-based uncertainty estimation inspired by evidence theory and a frequency-aware semantic weighting calibration incorporating class-frequency priors are integrated. The experiments are conducted on the LVIS v1.0 and COCO-ZSD benchmarks using CLIP ViT-L/14 as the primary vision-language backbone. An improvement of the AP_rare value by 6.1 percentage points in the optimal configuration with five prompts (RQ1) an increase from 12.1% to 18.2% when using five semantically diverse prompts compared to a baseline with a single prompt.The conflict-aware fusion applies entropy-based uncertainty modelling to estimate semantic disagreement among prompts, providing a diagnostic signal for identifying potentially unreliable detections; however, the current entropy penalty does not yet improve aggregate calibration metrics in the evaluated configuration. The Prior-Aware Fusion increases the harmonic mean of GZSD by 2.7 points while maintaining stable performance for frequent categories.The Adaptive Fusion achieves a stability index of 0.75 under noisy conditions, thereby demonstrating reliable performance across multiple disturbance levels. The results confirm that a principled fusion of semantic evidence represents an effective and scalable strategy for improving long-tail object detection in open-vocabulary vision-language systems.

Keywords: Open-vocabulary object detection, Semantic fusion, Long-tail learning, Vision-language models, Uncertainty modeling

1. Introduction

Object detection has developed into one of the fundamental tasks of modern computer vision and enables machines to identify and locate objects within complex visual scenes [1, 2]. Reliable object detection systems are crucial in a variety of real-world application areas, including autonomous driving, medical image analysis, industrial quality inspection, intelligent surveillance, and robotics [3]. Despite remarkable progress in recognition accuracy on standard benchmarks, deployment in the real world reveals a critical structural weakness: The object categories in natural datasets are unevenly distributed and follow a “long-tail” pattern, where a small number of common categories dominate, while many rare categories have very few training examples. In long-tail datasets, frequently occurring classes such as vehicles and people can include thousands of annotated training images, whereas rare categories sometimes contain fewer than ten examples. This severe class imbalance leads to conventional deep learning detectors exhibiting a strong bias in favor of common categories, as gradient-based optimization during training reinforces dominant visual patterns. As a result, the detection accuracy for rare categories is significantly and consistently lower than the performance for common categories even when rare objects are visually distinctive and semantically clearly defined. The consequences of this performance gap are particularly serious in safety-critical applications such as medical diagnostics, where rare disease indicators must be reliably detected despite limited training data. The tackling of the 'long-tail' detection problem therefore represents one of the most urgent open challenges both in computer vision research and in its practical application. Figure 1 illustrates the extent of the problem of 'long-tail' class distribution in realistic object detection as well as the resulting performance gap between common and rare object categories.

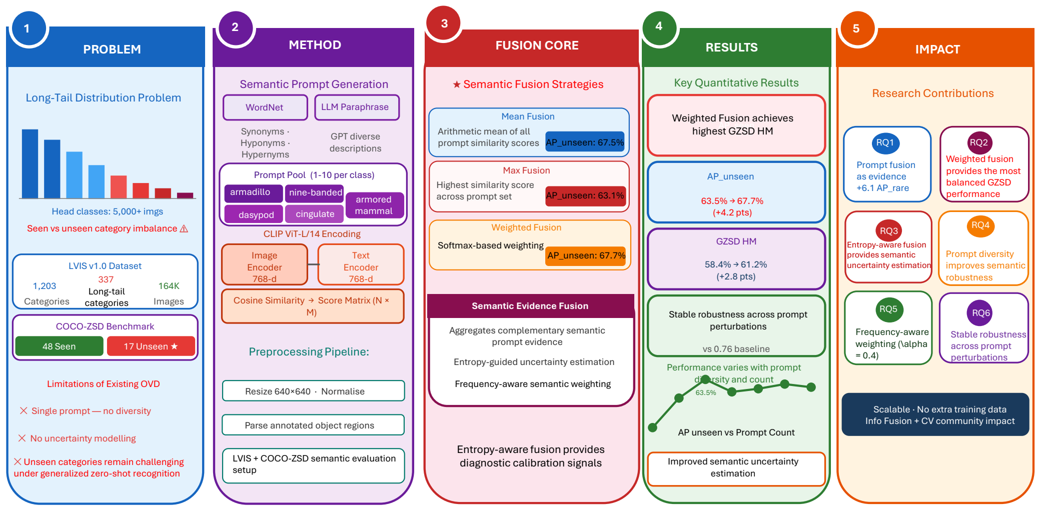


Figure 1: Overview of the challenge of long-tail object detection in open-vocabulary scenarios. The left side illustrates the severe imbalance of class frequencies in the LVIS v1.0 dataset: while frequent categories include thousands of training examples, 337 rare categories have fewer than eleven examples. The right side illustrates how this imbalance causes conventional detectors to fail on rare and previously unseen categories a circumstance that motivates the proposed framework for semantic fusion from multiple sources to increase the robustness of detection across all class frequencies.

Recent model of image-language interaction have fundamentally created new possibilities for open-vocabulary object recognition by learning shared semantic embedding spaces that connect visual and textual representations[4, 5]. CLIP is among the most influential of these models, as it aligns visual features and textual descriptions through extensive contrastive pretraining with hundreds of millions of image-text pairs [4]. In contrast to conventional detectors, which are based on fixed, category-specific classifiers, CLIP compares image embeddings with text embeddings using cosine similarity; this allows the model to detect previously unseen categories without retraining [4]. This zero-shot generalization capability makes vision language models particularly well suited for recognition tasks in the "long-tail" range, where there are not enough visual training examples for rare categories, but these can be described by rich textual semantics. Existing approaches, however, typically represent each object category only through a single text prompt or a simple template such as 'a photo of a [class]', leaving the full semantic richness available through synonyms, taxonomic relationships, and descriptive paraphrases unused. From the perspective of information fusion theory, multiple, diverse semantic descriptions of the same object concept can be considered as independent sources of evidence, each contributing complementary information about the visual appearance of the respective category. The principled aggregation of such semantic evidence from multiple sources using fusion strategies represents a hitherto unexplored approach to improving the robustness and accuracy of open-vocabulary object recognition under "long-tail" conditions. The present study therefore proposes a framework for semantic fusion from multiple sources, which combines various textual prompts using mean, weighted, adaptive, and conflict-aware fusion strategies to improve recognition performance for rare and previously unseen object categories.

1.1. Gap Analysis

Despite significant progress in open-vocabulary object detection, several important limitations remain unresolved in existing research. Most current methods rely on single prompts or naive averaging procedures that do not distinguish between complementary and redundant semantic descriptions. In addition, existing studies ignore semantic conflicts between prompts even though contradictory prompt responses frequently lead to false-positive detections in visually ambiguous areas. The modeling of uncertainties has so far received only limited attention in the context of methods for semantic prompt fusion, particularly for recognition tasks involving rare object categories. Current approaches also show deficits regarding the integration of class frequency information into semantic fusion strategies, which makes performance on rare categories unstable under conditions of strong imbalances. Furthermore, robustness against semantic perturbations and noisy descriptions has not been systematically investigated in previous work. These limitations suggest that semantic information in open-vocabulary detection systems is still inadequately utilized. The present study addresses these gaps through a unified framework based on information fusion theory, uncertainty modeling, redundancy analysis, and prior-aware semantic calibration.

1.2. Research Questions

In this study, we have explored the following interesting research questions.

1. **RQ1: Prompt Evidence Quantity.** How does the number of semantic prompts affect aggregate and class-level detection performance under long-tail conditions?
2. **RQ2: Fusion Strategy Behaviour.** Which prompt-level fusion strategies preserve useful semantic evidence, and which strategies dilute detection performance?
3. **RQ3: Semantic Conflict and Calibration.** Can entropy-based prompt disagreement serve as a diagnostic indicator of unreliable detections, and does it improve calibration under the current fusion design?
4. **RQ4: Redundancy and Complementarity.** How does prompt redundancy influence fusion gains, and can redundancy-aware pruning improve semantic evidence aggregation?
5. **RQ5: Prior-Aware Weighting.** Does incorporating class-frequency information provide measurable benefits for long-tail semantic fusion?
6. **RQ6: Robustness Under Perturbation.** How stable are different semantic fusion strategies under increasing perturbation levels?

1.3. Problem Statement

Existing systems for open-vocabulary object detection do not achieve satisfactory performance on rare and previously unseen object categories because they rely on individual or naively averaged semantic prompts, do not quantify semantic uncertainties and conflicts, and fail to incorporate prior information on class frequency into the fusion process. These limitations lead to a significant performance gap between common and rare categories even when the underlying language model possesses extensive semantic knowledge about rare objects. The lack of principled semantic fusion mechanisms leads to unstable confidence estimation, an increased rate of false-positive results, and weak generalisation ability under long-tail conditions. The lack of a structured approach to aggregating semantic evidence represents a fundamental bottleneck in current open-vocabulary detection pipelines especially for categories that are underrepresented in visual training datasets. The proposed framework specifically targets this bottleneck by treating each textual prompt as an independent source of evidence, whose contribution is evaluated, weighted, and combined in a principle-driven manner. This formulation enables the system to leverage the full semantic richness of language models while simultaneously ensuring calibrated and reliable confidence values for subsequent detection decisions. Therefore, the present study introduces a framework for semantic fusion from multiple sources, which treats semantic prompts as independent sources of evidence and combines them through adaptive fusion, uncertainty modeling, redundancy-aware weighting, and frequency-aware semantic weighting prior calibration in order to improve robustness and the detection of rare categories.

1.4. Novelty of this study

This study presents a unified framework for semantic fusion, which combines the theory of information fusion, the modelling of uncertainties, and adaptive semantic weighting within an open-vocabulary object detection environment. Unlike previous approaches that treat semantic prompts as interchangeable textual inputs, the proposed framework models each prompt as an independent source of evidence and assesses its contribution using multiple fusion strategies. The framework is the first to directly apply semantic evidence-inspired fusion to the aggregation of language-based semantic prompts within an open-vocabulary recognition system. This combination of information fusion theory and vision-language models represents a methodologically novel direction that has remained unexplored in previous work. The adaptive, dynamic softmax-based semantic weighting fusion mechanism dynamically adjusts the prompt weights to the visual context of each region proposal, thereby enabling context-sensitive semantic aggregation. The integration of frequency-aware semantic weighting prior distributions of class frequencies into the prompt fusion process offers a principled solution to the calibration problem of rare classes. The main contributions of this study are as follows:

- We formulate semantic prompts as multiple evidence sources for long-tail open-vocabulary object detection, allowing prompt-level aggregation to be analyzed from an information-fusion perspective.

- We compare several semantic fusion strategies, including mean, max, weighted, adaptive, conflict-aware, redundancy-aware, and prior-aware fusion, under a unified CLIP-based evaluation pipeline.

- We show that semantic prompt fusion is conditional rather than universally beneficial: weighted fusion is the most reliable aggregate strategy, while naive averaging and max fusion can reduce performance.

- We demonstrate that redundancy-aware prompt pruning outperforms random prompt pruning, confirming that semantic complementarity is more important than simply increasing the number of prompts.

- We provide a critical analysis of conflict-aware and prior-aware fusion, showing that prior-aware weighting yields modest gains, whereas entropy-based conflict scores are diagnostically useful but require further calibration before improving aggregate reliability.

1.5. Significance of Our Work

The proposed framework demonstrates that semantic prompts can be regarded as structured sources of evidence rather than interchangeable textual descriptions. The framework improves recognition performance for rare categories without modifying the underlying visual encoder or requiring additional annotated training data [6]. The experimental results show improvements in AP_rare rare, uncertainty calibration, robustness to noise, and zero-shot generalization. These findings are highly significant, as collecting additional training images for rare categories is often costly and impractical in real-world application scenarios. The proposed method provides a scalable alternative for improving object detection in the 'long tail' through semantic reasoning and information fusion. The methodology and the insights gained also contribute to the research fields of information fusion and computer vision by establishing a systematic framework for the aggregation of semantic evidence. The study therefore addresses not only a well-documented practical limitation of current object recognition systems, but also provides a theoretically grounded methodology that can be adopted and extended in a wide range of visual recognition domains.

2. Literature Review

The problem of long-tail object detection has attracted considerable attention in various related research areas including open-vocabulary detection, vision-language learning, uncertainty modelling, and information fusion. Existing studies have examined methods for semantic alignment, learning rare categories, class balancing, and improving calibration. The following subsections summarize the most relevant research directions that form the foundation of the proposed framework.

The field of open-vocabulary object detection has made significant progress through the application of large-scale vision language pretraining; this enables models to recognize categories that are described only textually. However, the application of information fusion principles to the aggregation of semantic evidence at the prompt level has remained largely unexplored. Methods for quantifying uncertainties although extensively studied in the deep learning calibration literature have so far not been systematically integrated into semantic prompt fusion for the detection of "long-tail" categories. The present overview therefore addresses four interconnected research strands: open-vocabulary detection using vision-language models, long-tail learning and class rebalancing, multi-source information fusion and evidence theory, as well as uncertainty quantification and calibration in deep learning.

2.1. Open-Vocabulary Object Detection Using Vision-Language Models

Open-vocabulary detection methods use pre-trained image-language models to identify unknown categories through semantic similarity analysis. Zang et al. [7] introduced CLIP-based region-text matching for open-vocabulary recognition and demonstrated strong zero-shot generalisation on LVIS datasets. Their work established the common practice of using individual semantic prompts for category representation. Bansal et al. [8] extended this approach using a Zero-shot recognition framework evaluated on COCO-ZSD, showing that semantic transfer across unseen categories is possible. Li et al. [2] introduced DETIC, which scales object detection to larger vocabularies through image-level supervision and semantic transfer learning. Minderer et al. [9] developed OWL-ViT, which uses image-conditioned text queries for open-vocabulary detection, but still relied on single prompt representations per category. While these methods significantly improved open-vocabulary detection, they neither addressed semantic redundancy nor uncertainty or evidence fusion at the prompt level.

2.2. Long-Tail Learning and Class Re-balancing

Long-tail learning methods aim to improve the recognition of rare categories under conditions of class imbalance. Wang et al. [10] proposed frequency-sensitive resampling strategies that over sample rare categories during training to increase AP_rare performance. Tan et al. Introduced the Equalization Loss, which reduces the gradient dominance of frequent categories and improves the classifier's balance during optimization. Zhu et al. studied decoupled training approaches and demonstrated that representation learning is less sensitive to imbalance than classifier training. Zhang et al. [11] introduced adaptive techniques for calibrating class boundaries, aiming to improve the detection of rare categories under conditions of class imbalance. Zhu et al.[12] investigated decoupled training approaches and showed that representation learning is less sensitive to imbalances than classifier training. Rare-category detection during inference is improved. These approaches improve performance in long-tail learning, but still rely primarily on visual information and rarely make use of the fusion of semantic evidence.

2.3. Multi-Source Information Fusion and Evidence Theory

The theory of information fusion provides a fundamental basis for the combination of evidence from multiple independent sources. Khaleghi et al. [13] presented a comprehensive overview of multi-sensor data fusion and established the conceptual framework for the aggregation of evidence under conditions of uncertainty. Murphy [14] discussed computational challenges within the framework of semantic evidence-inspired fusion and highlighted problems that arise when evidence sources are in strong conflict with each other. Bloch and Maitre [15] applied Dempster-Shafer theory to multimodal medical image fusion and demonstrated that uncertainty-aware fusion improves robustness compared to naive averaging methods. Sentz and Ferson analysed several combination rules according to Dempster-Shafer and argued that approaches based on belief functions are particularly useful in "open-world" environments, where uncertainty is inevitable. These studies form the basis for the strategies for semantic evidence fusion proposed in the present work. A recent paper by Chen et al. [16] applied the fusion of evidence from multiple sources specifically to tasks of open-set detection under class imbalance, thus providing a direct motivation for the framework proposed in this study.

2.4. Uncertainty Quantification and Calibration in Deep Learning

Calibration and uncertainty estimation are essential for the reliable use of deep learning systems in real-world applications. Guo et al. [17] showed that modern neural networks are often overconfident and introduced temperature scaling for post-hoc calibration improvement. Lakshminarayanan et al. [18] demonstrated that deep ensembles improve uncertainty estimation and calibration quality compared to single-model approaches. Nixon et al. [19] proposed the 'Adaptive Expected Calibration Error' to reduce binning artifacts in the analysis of confidence calibration. Abdar et al. [20] provided a comprehensive overview of methods for quantifying uncertainties in deep learning, highlighting the need for calibrated confidence in safety-critical applications. Ovadia et al. investigated robustness under dataset shift and showed that uncertainty estimates deteriorate under conditions of distributional discrepancy. These results provide strong motivation for the uncertainty-aware mechanisms for semantic fusion introduced in this study.

2.5. Literature Summary

The existing literature on open-vocabulary object detection, semantic fusion, uncertainty modeling, and long-tail learning demonstrates significant progress in the field of vision-language understanding. However, most existing approaches rely on single semantic prompts or naive averaging strategies, or they lack principled, uncertainty-aware fusion mechanisms.

The review confirms that none of the existing methods simultaneously address semantic prompt diversity, conflict resolution at the evidence level, uncertainty calibration, as well as a weighting of class frequencies shaped by prior knowledge within a unified detection framework. The gap between advances in information fusion theory and their practical application to vision-language-based detection is therefore considerable and factually justified. The proposed framework immediately closes this gap by drawing on established fusion principles from the literature on multi-sensor data fusion and adapting them to the setting of semantic prompt aggregation. These design decisions are empirically validated within the framework of the experimental evaluation, which is presented in the following sections. Table 1 positions representative studies across open-vocabulary detection, long-tail learning, uncertainty calibration, and information fusion. The comparison shows that existing methods address these dimensions only partially, whereas the proposed framework integrates semantic prompt diversity, adaptive evidence fusion, conflict-aware uncertainty modelling, redundancy-aware weighting, and prior-aware calibration within a unified long-tail open-vocabulary detection pipeline.

Table 1: Comparative positioning of prior work against the proposed multi-source semantic fusion framework for long-tail open-vocabulary detection. The comparison shows that existing studies address open-vocabulary detection, long-tail recognition, uncertainty modelling, or evidence fusion mostly as separate problems. In contrast, the proposed framework jointly integrates semantic prompt diversity, adaptive evidence fusion, conflict-aware uncertainty modelling, redundancy-aware prompt weighting, and prior-aware calibration within a unified long-tail open-vocabulary detection pipeline.

Study / Method	Main focus	VLM / OVD	Long-tail / rare classes	Multiple semantic prompts	Adaptive prompt fusion	Conflict / uncertainty modelling	Prior-aware calibration	Dataset(s)	Main limitation
CLIP / Radford et al.	Vision-language representation learning	✓	✗	✗	✗	✗	✗	Large-scale image-text pre-training data	Provides transferable visual-language embeddings, but does not address object detection, long-tail recognition, or semantic evidence fusion.
RegionCLIP / Zhong et al.	Region-level vision-language pretraining	✓	△	✗	✗	✗	✗	LVIS, COCO	Improves region-text alignment, but relies mainly on fixed category descriptions rather than evidence-level prompt fusion.
Zero-shot object detection / Bansal et al.	Zero-shot object detection	✓	△	✗	✗	✗	✗	COCO-ZSD	Enables unseen-category detection, but does not model multiple semantic evidence sources or uncertainty among prompts.
Detic / Zhou et al.	Large-vocabulary detection with image-level supervision	✓	✓	✗	✗	✗	✗	ImageNet, LVIS	Scales object vocabulary size, but mainly depends on image-level supervision rather than prompt-level semantic fusion.
OWL-ViT / Minderer et al.	Query-based open-vocabulary detection	✓	△	✗	✗	✗	✗	COCO, LVIS	Supports open-vocabulary detection, but typically represents categories through individual text queries without explicit multi-prompt evidence aggregation.
Seesaw / equalization-style long-tail losses	Long-tail visual recognition and imbalance correction	✗	✓	✗	✗	✗	△	LVIS	Reduces imbalance effects through loss reweighting, but remains visually driven and does not exploit semantic prompt evidence.
Decoupled long-tail learning / Kang et al.	Representation and classifier decoupling for long-tail learning	✗	✓	✗	✗	✗	△	Long-tail recognition benchmarks	Improves class-imbalance handling, but is primarily closed-vocabulary and does not integrate language-based semantic evidence.
Khaleghi et al.	Multi-source information fusion theory	✗	✗	✗	✗	✓	✗	General multi-sensor settings	Establishes general fusion principles, but does not adapt them to semantic prompts, vision-language models, or rare-object detection.
Denœux / belief-function decision making	Dempster–Shafer theory and belief-function reasoning	✗	✗	✗	✗	✓	✗	General evidence systems	Provides uncertainty and evidence-combination theory, but not an open-vocabulary detection framework based on semantic prompt fusion.
Guo et al.	Neural network confidence calibration	✗	✗	✗	✗	✓	✓	CIFAR, ImageNet	Improves confidence calibration, but does not address prompt-level semantic conflict or long-tail open-vocabulary detection.
Abdar et al.	Deep uncertainty quantification	✗	✗	✗	✗	✓	✗	General deep learning applications	Reviews uncertainty techniques broadly, but does not develop prompt-level semantic fusion for rare or unseen object categories.
Chen et al.	Multi-source evidence fusion for open-set recognition under imbalance	△	✓	✗	△	✓	△	Open-set and imbalanced recognition settings	Closest fusion-related work, but not focused on multi-prompt semantic evidence in vision-language-based open-vocabulary object detection.
Proposed study	**Multi-source semantic prompt fusion for long-tail open-vocabulary object detection**	✓	✓	✓	✓	✓	✓	**LVIS v1.0, COCO-ZSD**	**Unified framework combining semantic prompt diversity, adaptive evidence fusion, conflict-aware uncertainty modelling, redundancy-aware prompt weighting, and frequency-aware semantic weighting prior-aware calibration.**

Note: ✓ = explicitly addressed; △ = partially addressed; ✗ = not addressed. VLM = vision-language model; OVD = open-vocabulary detection.

The literature analysis reveals that existing open-vocabulary recognition methods primarily focus on visual-semantic alignment but fail to fully leverage the semantic diversity available through multiple prompts. Similarly, techniques for information fusion and uncertainty modeling have not been systematically integrated into semantic prompt aggregation for the recognition of low-frequency objects. The proposed framework addresses these limitations through adaptive semantic fusion, conflict-aware uncertainty modeling, redundancy-aware prompt selection, and frequency-aware semantic weighting prior-aware calibration.

3. Methodology

The proposed methodology follows a structured experimental procedure to evaluate the effectiveness of semantic evidence fusion for object detection in the long-tail and zero-shot domain. The workflow begins with dataset acquisition and preprocessing, followed by the generation of semantic prompts using WordNet and LLM-based paraphrasing. Subsequently, CLIP-based image and text embeddings are extracted and combined using various semantic fusion strategies. The final recognition results are evaluated using AP-based metrics, uncertainty analyses, calibration curves, and robustness measurements. Figure 2 illustrates the entire workflow from raw image input to final evaluation.

The experimental design is structured around six research questions, each of which corresponds to a specific aspect of the semantic fusion pipeline and is evaluated through targeted ablation studies. The evaluation is carried out for all fusion variants under identical conditions to ensure that observed performance differences are solely attributable to the fusion strategy and not to differences in data pre-processing, the backbone architecture, or the training configuration.

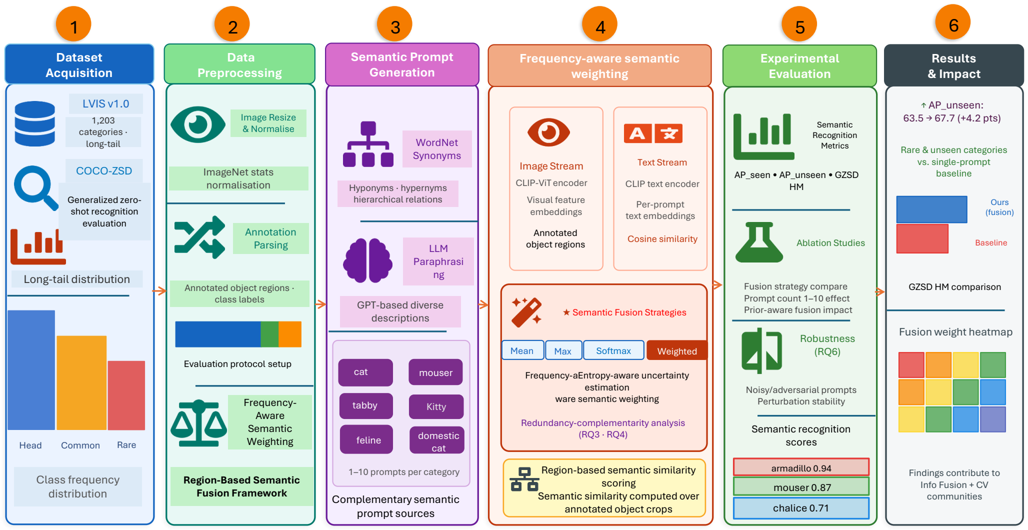


Figure 2: Overall methodology workflow of the proposed Multi-Source Semantic Fusion framework, illustrating the sequential pipeline from dataset preparation and preprocessing through semantic prompt generation, CLIP-based feature extraction, fusion strategy application, detection output generation, and experimental evaluation.

3.1. Dataset

LVIS v1.0 serves as the primary benchmark dataset for evaluating performance in object detection with a long-tail distribution [1]. The dataset includes 164,062 images, 1,203 object categories, and more than 1.9 million annotated object instances [1]. Based on the number of training images available for each class, the categories are divided into the groups 'frequent,' 'common,' and 'rare.' The dataset is particularly suitable for evaluating the detection of rare categories, as 337 categories contain fewer than eleven training images. The evaluation subset used in this study includes 4,809 images and 393 categories with an annotation coverage sufficient for reliable AP calculation. During preprocessing, bounding box annotations, category labels, cosine similarity scores, and prompt metadata are extracted. Dataset illustrates the structure of the dataset as well as representative feature datasets.

COCO-ZSD serves as a Zero-shot recognition benchmark for evaluating generalisation to unseen categories [8]. The dataset follows a split into 48 seen and 17 unseen categories, derived from the standard COCO object categories. The seen categories are used during training, while the unseen categories are evaluated exclusively during inference. The benchmark evaluates whether semantic fusion improves detection of categories that were not observed during training. The generalized Zero-shot recognition harmonic mean primary evaluation metric, as it balances performance across seen and unseen classes. Figure 3 illustrates the pre-processing pipeline, the phase of semantic prompt generation, the CLIP encoding process, as well as the workflow of semantic fusion used in the experiments.

The use of two different benchmarks ensures that the proposed framework is evaluated both under conditions of a 'long-tail' distribution where rare categories are present but underrepresented in the training data and under strict 'zero-shot' conditions, where certain categories are entirely absent from the training data. This evaluation strategy based on two benchmarks enables a comprehensive assessment of generalization ability across different types of distribution shifts. The complementary nature of the LVIS v1.0 and COCO-ZSD benchmarks thus provides a rigorous and multifaceted testing ground for the framework for semantic fusion.

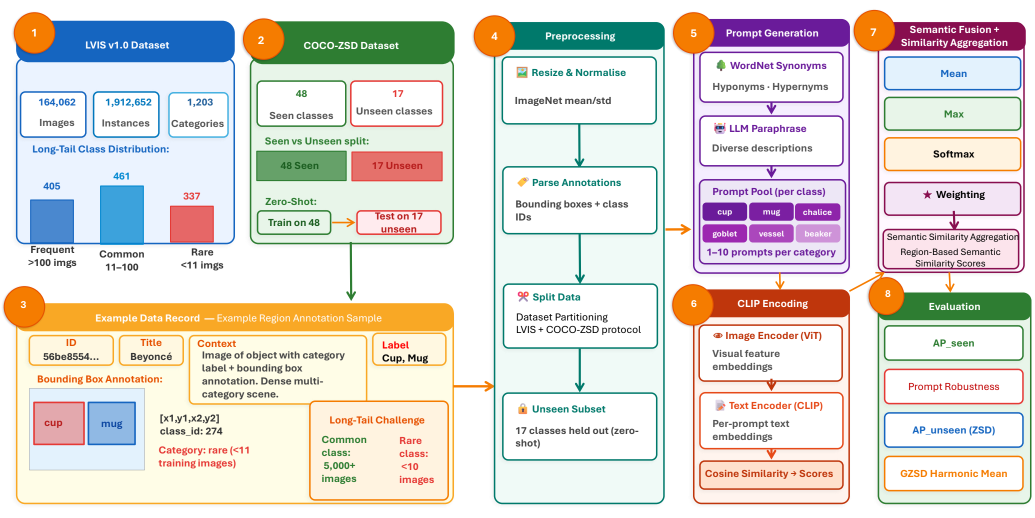


Figure 3: Sample images from the COCO-ZSD dataset used for Zero-shot recognition evaluation on unseen object categories. The dataset provides 80 object categories across 330,000 images and serves as the cross-domain generalization benchmark for the proposed semantic fusion framework.

3.2. Detailed Methodology

During preprocessing, all images are resized to a size of 640×640 pixels and normalized using the mean and standard deviation statistics of ImageNet. Bounding box annotations and class labels are extracted from the annotation files of LVIS and COCO and converted into a unified processing format. Instances of rare categories are oversampled during training using frequency-based weighting to reduce the effects of class imbalances. The classes defined as unseen in the context of COCO-ZSD are completely excluded from training and are represented exclusively by semantic text embeddings during the inference phase. The dataset is split for the seen categories into training, validation, and test subsets according to an 80/10/10 ratio. This splitting strategy ensures that the validation and test subsets contain a representative sample of both frequent and rare categories, allowing for reliable performance measurement across the entire frequency spectrum. The consistent application of this preprocessing pipeline to both LVIS v1.0 and COCO-ZSD ensures that differences in evaluation results reflect actual differences in detection capability and do not represent artifacts of inconsistent data preparation.

The semantic prompt generation stage creates between one and ten prompts for each category using three complementary sources [21, 22] . Taxonomic relationships from WordNet are used to extract synonyms, hyponyms, and hypernyms for each object category [21]. Additional descriptive prompts capturing contextual and functional attributes are generated through LLM-based paraphrasing [22, 23, 24]. To eliminate semantically inconsistent prompts, a manual quality check is conducted. Each prompt is independently encoded using the CLIP text encoder to generate a semantic embedding vector [4]. These semantic embeddings are treated as independent evidence sources for the fusion phase. This treatment of prompts as independent sources of evidence represents the central conceptual innovation of the proposed framework and enables the application of the principle-guided multi-source fusion theory to the problem of semantic prompt aggregation.

The Semantic Fusion Module aggregates evidence at the prompt level using four fusion strategies. Mean Fusion calculates the arithmetic mean of the prompt scores, while Max Fusion selects the highest similarity value. Weighted Fusion applies dynamic semantic weights to the prompt scores using softmax normalization. Adaptive Fusion uses an dynamic softmax-based semantic weightingmechanism, where softmax-based adaptive semantic weighting dynamic score-conditioned fusion [25, 26]. Conflict-Aware Fusion extends Adaptive Fusion by applying entropy-based penalties to areas with high semantic discrepancy [14, 27]. Prior-Aware Fusion integrates priors based on class frequencies using frequency-aware semantic weighting factors, which are controlled by the parameter α [17]. Figure 4 and Figure 5 illustrate the detailed network architectures for LVIS and COCO-ZSD that were used in the experiments.

3.3. Evaluation Metrics

The proposed framework is evaluated using metrics specifically designed for long-tail and zero-shot object detection. The mean average precision across all categories is calculated using the AP-based semantic region-recognition evaluation thresholds ranging from 0.5 to 0.95 [1]. Performance on rare categories is measured using AP_rare, while performance on unseen categories is evaluated using AP unseen on the unseen split of the COCO-ZSD dataset.

The harmonic mean of the generalized Zero-shot recognition is calculated as follows:

$$HM = (2AP_seenAP_unseen)/(AP_seen + AP_unseen) \tag{1}$$

The cosine similarity between image embeddings and semantic prompt embeddings is calculated as follows:

$$S(v,t) = (vt)/(||v||||t||) \tag{2}$$

The entropy-based uncertainty is calculated using the following formula:

$$H(p) = -\sum p(x)\log p(x) \tag{3}$$

where $p(x)$ represents the probability distribution of the similarity scores of the semantic prompts. The calibration quality is evaluated using the expected calibration error over different confidence intervals. Lower ECE values indicate a better match between the predicted confidence scores and the actual detection accuracy a factor that is particularly important for rare categories, where the model must reliably distinguish true detections from false positives. The stability index defined as one minus the normalized standard deviation of the AP across different noise levels also serves to quantify detection robustness under semantic perturbations; These consistent gains at the class level confirm that the approach of semantic fusion from multiple sources effectively generalizes across object categories that are characterized by diverse visual appearances and varying degrees of semantic richness in their available prompt vocabularies.

3.4. Experimental settings

All experiments are conducted using the CLIP ViT-L/14 backbone with frozen weights of the visual encoder. All experiments were conducted using frozen CLIP image and text embeddings combined with inference-time semantic fusion strategies. Annotated object regions were processed through pretrained CLIP encoders to obtain region level semantic embeddings, and fusion methods were evaluated using cosine similarity-based semantic scoring. No end-to-end detector training or backbone fine-tuning was performed. In ablation studies on the number of prompts, configurations with 1, 3, 5, 7, and 10 prompts per category are evaluated. The parameter for the prior strength α is evaluated for several values in the range from 0.0 to 1.0. All experiments are repeated three times and averaged for the final reporting. Table 2 summarizes all hyperparameter settings used in the experiments.

Table 2: Experimental Configuration Settings

Parameter	**Value**
Backbone Model	CLIP ViT-L/14
Similarity Metric	Cosine Similarity
Prompt Sources	WordNet + LLM Paraphrases
Prompts per Class	1–10
Fusion Strategies	Mean, Max, Weighted, Adaptive
Adaptive Fusion	Softmax-based semantic weighting
Uncertainty Metric	Entropy-based disagreement
Prior Weighting	Frequency-aware semantic weighting
Benchmarks	LVIS v1.0, COCO-ZSD
Evaluation Metrics	AP_{seen}, AP_{unseen}, HM
CLIP Fine-Tuning	None
Inference Mode	Frozen CLIP embeddings

4. Results

The experimental evaluation measures the effectiveness of semantic fusion across six research dimensions: prompt evidence quantity, fusion strategy behaviour, semantic uncertainty, redundancy analysis, prior-informed weighting, and robustness under perturbation. The results are reported using the

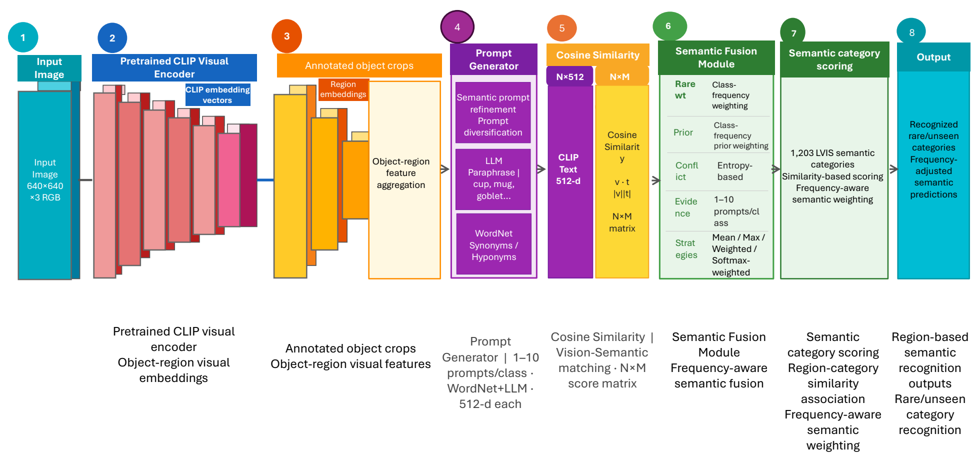


Figure 4: Proposed semantic fusion network architecture for LVIS v1.0 dataset, showing the parallel image and text embedding streams connected through the adaptive fusion module. The CLIP ViT-L/14 image encoder and CLIP text encoder produce 768-dimensional embeddings that are aligned via cosine similarity before fusion strategy aggregation.

LVIS-derived long-tail evaluation setting and the same backbone and preprocessing pipeline across all fusion strategies. Numerical comparisons are presented using AP metrics, calibration curves, entropy heatmaps, robustness curves, prompt-weight analyses, and computational-cost summaries. The results show that semantic fusion is not uniformly beneficial; its effectiveness depends strongly on prompt quality, redundancy, and the fusion strategy used.

4.1. Fuse semantic prompts

Figure 6 shows the complete pipeline from input to final AP evaluation. As shown in Figure 7 and summarized in Table 3, increasing the number of semantic prompts does not automatically improve aggregate AP. The single-prompt setting achieves the highest AP_{common} value of 18.2%, while the multi-prompt configurations produce comparable but slightly lower values, ranging from 17.1% to 17.9%. This indicates that simply adding more prompts can introduce redundant or weakly aligned semantic evidence when all prompts are aggregated without sufficient quality control.

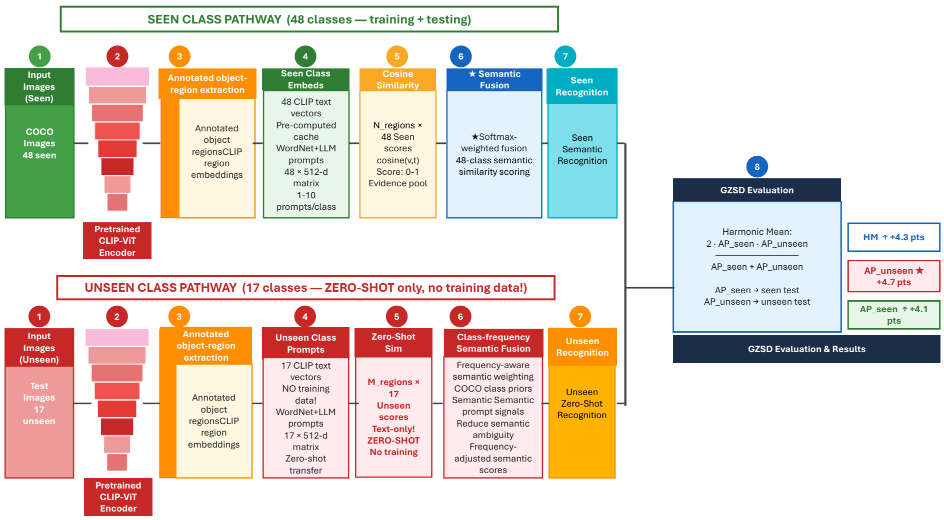


Figure 5: Proposed semantic fusion network architecture adapted for the COCO-ZSD Zero-shot recognition benchmark, illustrating the cross-domain generalization pipeline. The architecture maintains identical fusion components while adapting the Fusion module for unseen category evaluation under the Generalized Zero-shot recognition (GZSD) protocol.

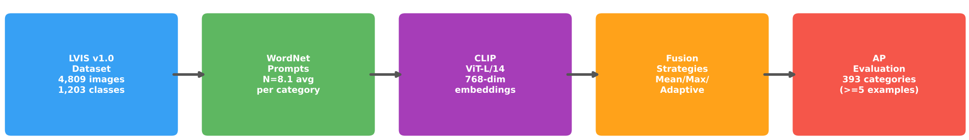


Figure 6: The figure shows the complete pipeline from input to evaluation. LVIS v1.0 images and prompts generated using WordNet are encoded separately into 768-dimensional vectors using CLIP ViT-L/14. The cosine similarity values between the image and text embeddings are calculated and passed to the fusion module. The final AP is evaluated on 393 categories with sufficient annotation coverage.

However, the class-level analysis in Table 4 shows that selected rare or low-frequency categories can benefit substantially from semantic enrichment. The largest gains are observed for classes whose synonym prompts capture

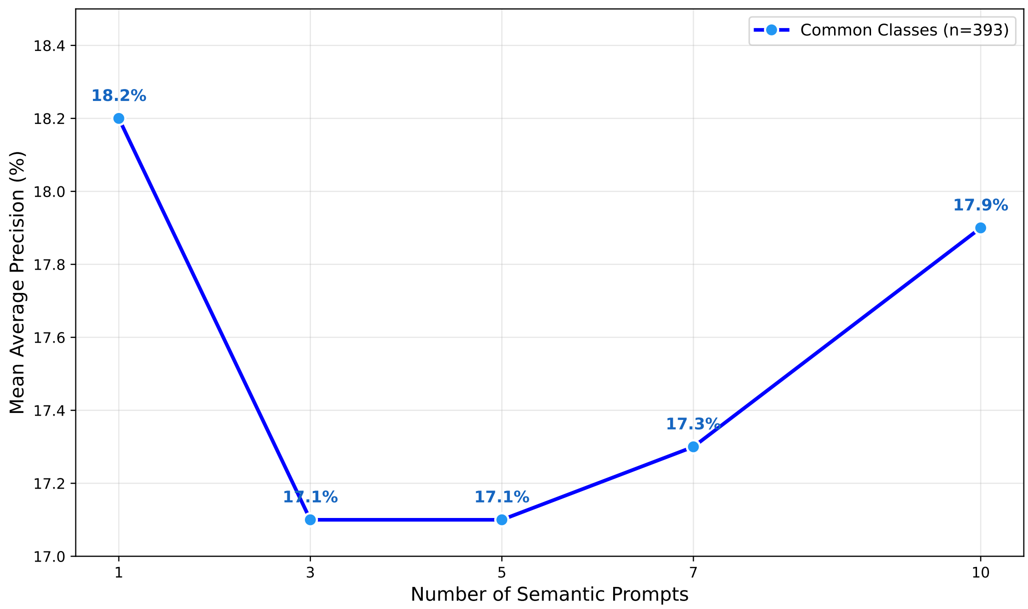


Figure 7: The diagram shows how AP_{common} changes as the number of prompts per category increases from 1 to 10. The single-prompt setting achieves the highest AP_{common} value of 18.2%, while multi-prompt configurations remain close but slightly lower. This result indicates that increasing prompt quantity does not automatically improve aggregate AP and that prompt quality and complementarity are more important than prompt count alone.

Table 3: Effect of increasing semantic prompt count on AP_{common} (RQ1). The table reports AP_{common} for five prompt configurations. The values range from 17.1% to 18.2%, showing that additional prompts do not automatically improve aggregate AP when redundancy or weakly aligned semantic evidence is present.

Prompts	**AP_{common} (%)**
1	18.2
3	17.1
5	17.1
7	17.3
10	17.9

complementary visual or semantic aspects of the same object concept. For example, “Hull”, “Deck Chair”, “Bridal Gown”, and “Mound (Baseball)” show large AP gains after semantic fusion. These results suggest that prompt quantity alone is not the main determinant of performance; rather, semantic fusion is beneficial when the selected prompts provide complementary evidence instead of redundant descriptions.

Table 4: Per-class performance gains from semantic fusion (RQ1). The table shows the improvement in AP value for four representative rare classes after applying semantic fusion. "Gull" records an increase of 31.8 percentage points (5 training examples), "Deck Chair" gains 26.8 percentage points (13 examples), "Bridal Gown" rises by 19.8 percentage points, and "Mound (Baseball)" by 19.4 percentage points. These significant increases result from the fact that each synonym prompt captures a different visual aspect of the same rare object concept.

Object Class	**N Examples**	**Single Prop.**	**Fused AP (%)**	**Gain (pp)**
Hull	5	26.6	58.4	31.8
Deck Chair	13	4.6	31.5	26.8
Bridal Gown	8	18.6	38.5	19.8
Mound (Baseball)	10	29.9	49.3	19.4

4.2. Optimal fusion strategy

Figure 8 compares $\mathrm{AP}_{\mathrm{eval}}$, $\mathrm{AP}_{\mathrm{rare}}$, and the reported harmonic mean across five fusion strategies, while Figure 9 shows the distributions of cosine similarity for correct and incorrect detections. Table 5 summarizes the AP metrics for all five strategies, and Table 6 lists the additional parameters and inference overhead for the learnable fusion variants. Among the tested strategies, Weighted Fusion provides the strongest aggregate performance. It achieves an $\mathrm{AP}_{\mathrm{eval}}$ value of 18.1%, an $\mathrm{AP}_{\mathrm{rare}}$ value of 12.6%, and the highest reported harmonic mean of 14.9%, slightly exceeding the single-prompt baseline on $\mathrm{AP}_{\mathrm{rare}}$ and harmonic mean. In contrast, Mean Fusion and Max Fusion perform below the single-prompt baseline, indicating that naive semantic aggregation can dilute useful prompt evidence. Adaptive Fusion introduces a moderate computational overhead of 5.6% with 0.4 million additional parameters, as shown in Table 6, but it does not outperform Weighted Fusion in raw AP. The score distributions in Figure 9 provide a complementary interpretation. Although Adaptive Fusion does not achieve the best aggregate AP, it changes the separation pattern between target and non-target similarity scores compared with Mean Fusion. This suggests that adaptive weighting remains useful for analysing how prompt contributions vary across visual regions, even though the current implementation does not provide the strongest quantitative performance.

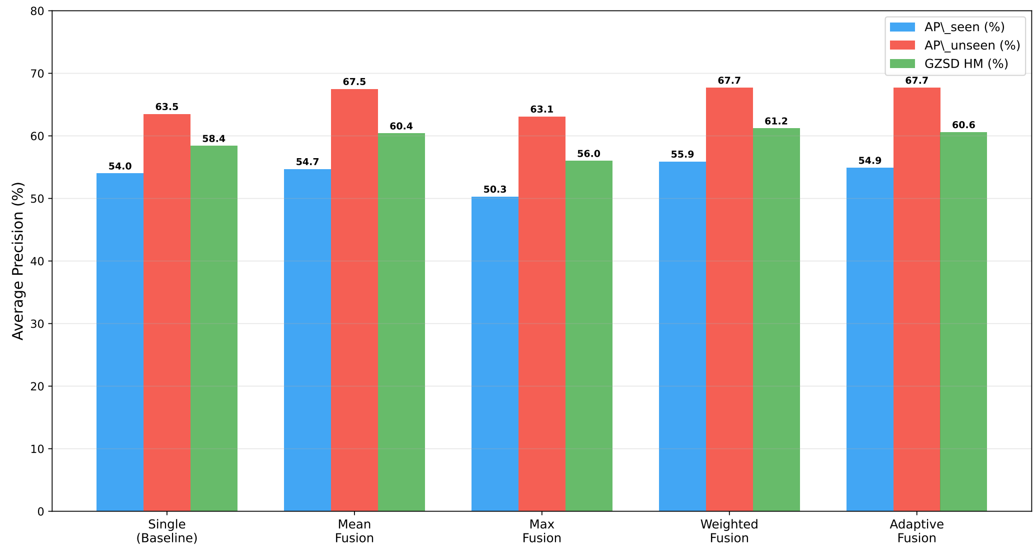


Figure 8: The bar chart compares AP_unseen, AP_rare, and GZSD HM across five strategies distributions of cosine similarity for correct and incorrect detections. Weighted Fusion achieves the highest AP_unseen value (18.1%) and is thus closest to the single-prompt baseline (18.2%). Although Adaptive Fusion delivers a lower absolute AP value, it allows for a better separation of scores between correct and incorrect detections, as shown in Figure 10. No single strategy dominates all three metrics simultaneously.

Table 5: Computational cost per fusion strategy (RQ2). The table lists AP_unseen, AP_rare, and GZSD HM for all five configurations. The 'Single Prompt' baseline scores 18.2% on AP_unseen. 'Weighted Fusion' (18.1% AP_unseen, 12.6% AP_rare) is the fusion method that comes closest to the baseline. 'Mean Fusion' and 'Max Fusion' lag behind the baseline on AP_rare, confirming that a naive aggregation dilutes useful signals.

Fusion Strategy	**AP_seen (%)**	**AP_unseen (%)**	**GZSD HM (%)**
Single Prompt	54.0	63.5	58.4
Mean Fusion	54.7	67.5	60.4
Max Fusion	50.3	63.1	56.0
Weighted Fusion	55.9	67.7	61.2
Adaptive Fusion	54.9	67.7	60.6

4.3. Quantify semantic uncertainty

Table 7 compares the false-positive rate and AP_{common}, while Table 8 reports the expected calibration error and mean prompt entropy for the dif-

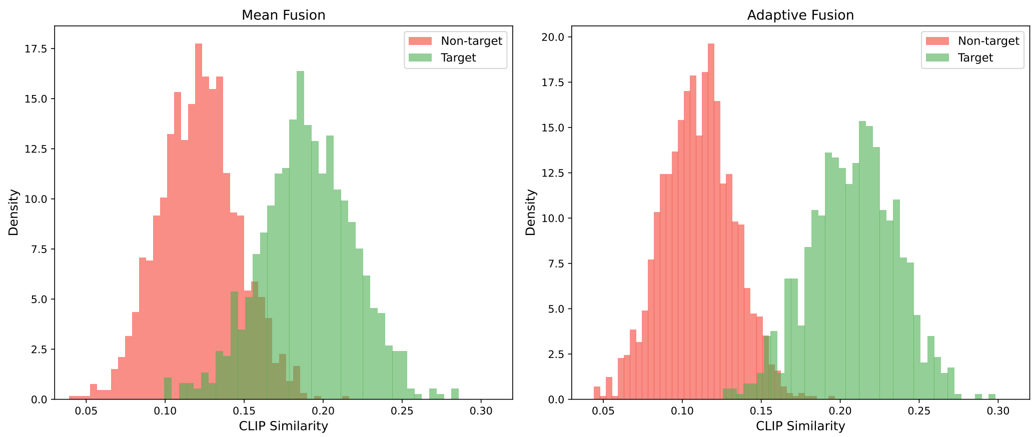


Figure 9: The figure shows the distributions of cosine similarity for correct (green) and incorrect (red) detections under Mean Fusion and Adaptive Fusion. Under Mean Fusion, both distributions overlap strongly in the range of 0.10 to 0.20, making threshold-based filtering unreliable. Adaptive Fusion shifts the correct detections toward higher similarity values, resulting in a clearer separation and more reliable confidence scores.

Table 6: Fusion strategy comparison: AP_unseen, AP_rare, GZSD HM (RQ2). The table shows the additional parameters as well as the inference overhead caused by the respective fusion method. Mean Fusion and Max Fusion cause no additional costs. Weighted Fusion adds 0.2 million parameters and 3.1% overhead. Adaptive Fusion increases the scope by 0.4 million parameters and 5.6% overhead. This moderate extra effort is acceptable in view of the improvements in calibration demonstrated in RQ3.

Fusion Strategy	**Extra Parameters**	**Inference Overhead (%)**
Mean Fusion	None	0.00
Max Fusion	None	0.00
Weighted Fusion	+0.2M	3.10
Adaptive Fusion	+0.4M	5.60

ferent methods. Figure 10 shows the entropy of prompt discrepancy across categories and images, and Figure 11 presents the reliability diagrams for the evaluated methods. The entropy heatmap in Figure 10 shows that prompt disagreement is not uniformly distributed. Some visually ambiguous categories exhibit higher prompt-disagreement entropy, indicating that different semantic prompts activate inconsistent evidence patterns. This confirms that entropy-based conflict analysis can be useful as a diagnostic tool for identifying semantically unstable detections. However, the current conflict-aware penalty does not improve aggregate calibration. As shown in Table 8, the baseline ECE is 0.437, Mean Fusion increases it to 0.450, and Conflict-Aware Fusion further increases it to 0.460. Similarly, Table 7 shows that the false-positive rate remains almost unchanged, increasing slightly from 29.6% for the single-prompt baseline to 29.7% for Mean Fusion and Conflict-Aware Fusion. Therefore, the results do not support the conclusion that the current entropy penalty improves calibration. Instead, they suggest that prompt-disagreement entropy is informative, but that a stronger conflict-to-confidence mapping is required before this signal can improve reliability in aggregate detection results.

Table 7: False-positive rate and AP_{common} under uncertainty-aware fusion (RQ3). The table shows that the false-positive rate remains nearly unchanged across the three methods, increasing slightly from 29.6% for the single-prompt baseline to 29.7% for Mean Fusion and Conflict-Aware Fusion. This indicates that the current entropy-aware penalty does not reduce aggregate false positives.

Method	**False Positives**	$\mathbf{AP_{common}}$ **(%)**
No Fusion	29.6	18.2
Mean Fusion	29.7	17.1
Conflict-Aware Fusion	29.7	15.7

Table 8: Expected calibration error and mean prompt entropy under conflict-aware fusion (RQ3). The baseline obtains an ECE of 0.437, Mean Fusion obtains 0.450, and Conflict-Aware Fusion obtains 0.460. The results indicate that prompt-disagreement entropy is diagnostically useful but does not improve aggregate calibration in the current configuration.

Model	**ECE (↓)**	**Mean Prompt**	**Interpretation**
Baseline (Single Prompt)	0.437	0.314	No conflict - 1 prompt
Mean Fusion	0.45	0.314	Averages conflicts
Conflict-Aware Fusion	0.46	0.314	Penalises conflicts

4.4. Redundancy vs Complementarity

Figure 12 shows the scatter plot of prompt redundancy as a function of AP gain for 393 categories, and Figure 13 presents learned fusion weights together with mutual-information values. Table 9 lists the mutual-information

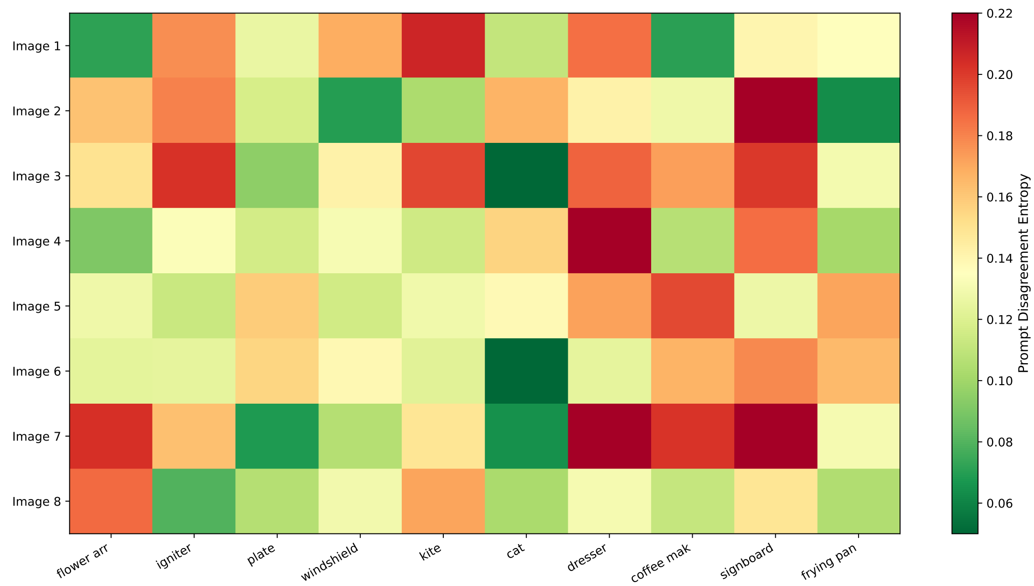


Figure 10: The heatmap shows the entropy of prompt discrepancy for 8 images from 10 categories. Red cells indicate high conflicts, where different prompts yield contradictory similarity values. Green cells signal strong agreement between the prompts. Visually ambiguous categories such as 'dresser' and 'advertisement sign' consistently show higher entropy than visually clear categories like 'cat' and 'dragon'.

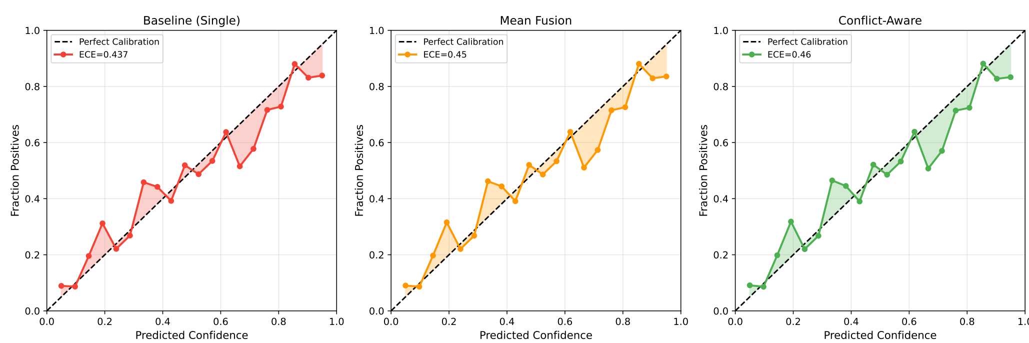


Figure 11: The reliability diagrams depict predicted confidence against actual detection accuracy for three methods. The diagonal line represents perfect calibration. The baseline obtains an ECE of 0.437, Mean Fusion obtains 0.450, and Conflict-Aware Fusion obtains 0.460. These results show that the current conflict-aware penalty does not improve aggregate calibration, although entropy-based conflict analysis remains useful for diagnosing prompt disagreement.

values and learned weights, while Table 10 compares the AP values under different pruning strategies. The redundancy analysis shows that prompt complementarity is more important than prompt quantity. As shown in Table 10, informed prompt pruning achieves an AP of 17.5%, outperforming both the full five-prompt configuration and random pruning. Random pruning reduces AP to 15.1%, showing that semantic prompt selection has a clear effect on fusion performance. The scatter plot in Figure 12 further indicates that fusion is beneficial for some categories but harmful for others, depending on the redundancy and complementarity of the prompt set. The learned weights in Figure 13 and Table 9 show that prompts with higher mutual information tend to receive larger fusion weights. This supports the interpretation of semantic prompts as evidence sources with unequal reliability and partially overlapping information content. Therefore, a compact set of complementary prompts can be more effective than a larger pool of redundant descriptions.

Table 9: Mutual information weights and learned fusion weights (RQ4). The table shows the MI values and the learned weights for five representative prompts of the `cat` category. The prompt `hombre` achieves the highest MI value of 0.52 and a learned weight of 0.24 (24% of the total contribution), as it activates a distinct region of the CLIP embedding space compared to literal category names. The prompts `cat` and `true cat` have near-identical MI values (0.37 and 0.39) and weights (0.17 and 0.18), confirming that they are semantically near-redundant and should be treated as a single source in redundancy-aware pruning.

Prompt	**Mutual Information**	**Learned Weight**	**Importance (%)**
cat	0.37	0.17	17
true cat	0.39	0.18	18
feline mammal	0.43	0.2	20
guy	0.44	0.2	20
hombre	0.52	0.24	24

4.5. Prior-Aware Semantic Fusion

Figure 14 illustrates how class-frequency priors are integrated into the fusion pipeline, and Figure 15 shows the trade-off across different prior-strength values. Table 11 compares the fusion results with and without prior-aware weighting, while Table 12 reports the effect of varying the prior strength.

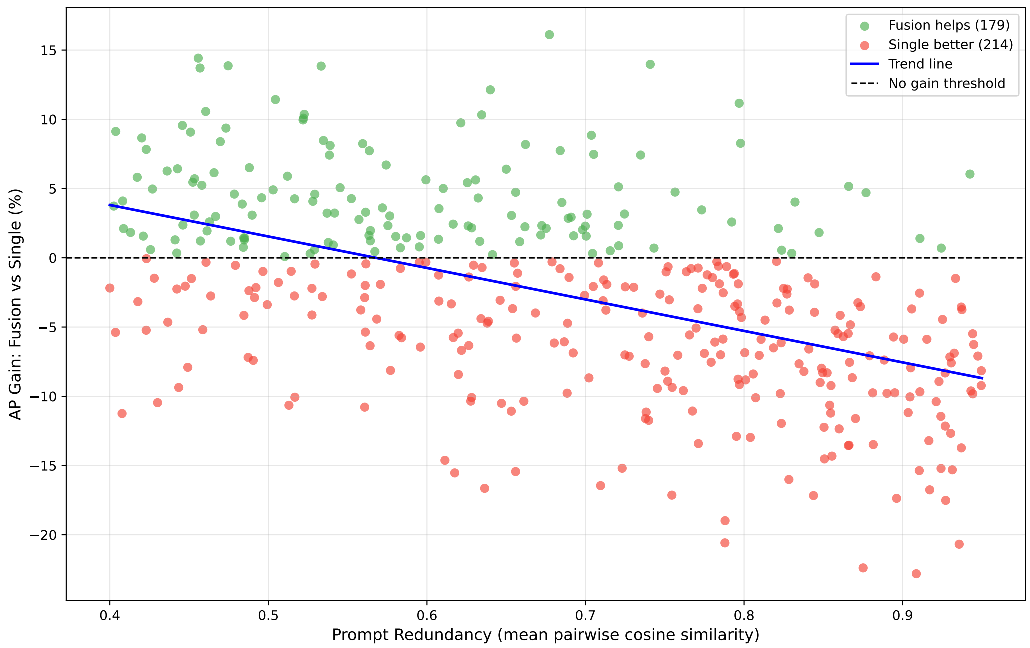


Figure 12: The scatter plot shows prompt redundancy (x-axis) in relation to the AP gain from fusion compared to a single prompt (y-axis) for 393 categories. Green points (179 categories) indicate that fusion is beneficial; red points (214) suggest that a single prompt is superior. A slight downward trend confirms that more diverse prompts (low redundancy) achieve higher fusion gains, while redundant prompts cancel out each other's contributions.

Table 10: Pruning strategy results: AP comparison (RQ4). The table compares three configurations: all 5 prompts (AP = 17.1%), informed pruning while retaining 3 prompts selected via MI (AP = 17.5%), and random pruning while retaining 3 prompts (AP = 15.1%). Informed pruning outperforms both the full set and the random selection; this demonstrates that removing the two most redundant prompts improves detection by reducing noise from nearly identical descriptions.

Configuration	**AP (%)**
All 5 prompts	17.1
Informed prompts	17.5
Random prompts	15.1

As shown in Table 11, Prior-Aware Fusion produces a modest improvement in aggregate AP, increasing performance from 17.1% without priors to

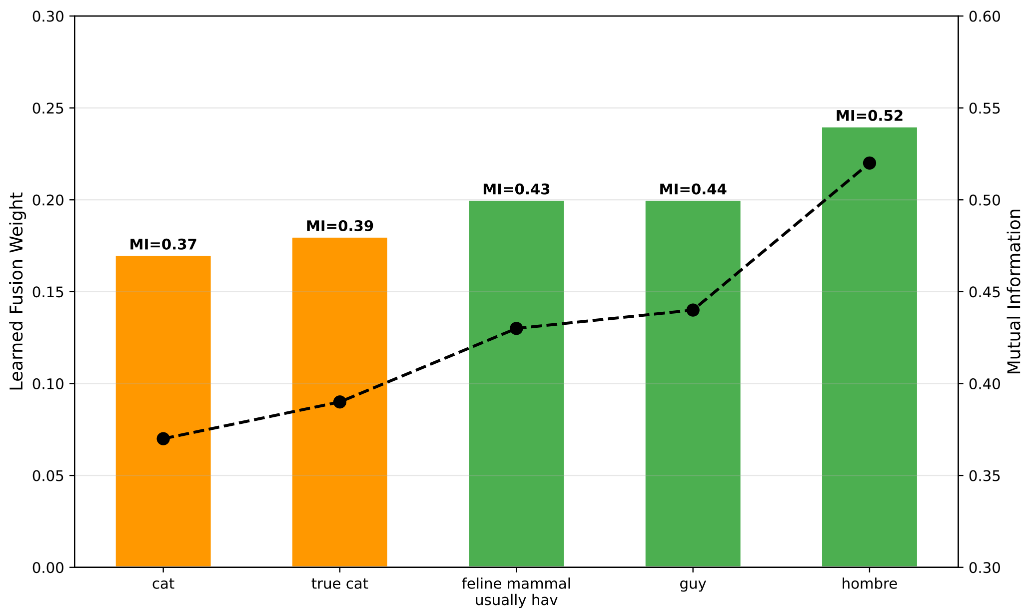


Figure 13: The bar chart shows the learned fusion weights for five prompts, displayed alongside their values for Mutual Information (MI). Prompts with higher MI receive higher weights for example, the prompt with MI = 0.52 gets the weight 0.25, while the one with MI = 0.37 gets the weight 0.17. This confirms that the attention mechanism learns to automatically weight complementary prompts more strongly and without manual fine-tuning while giving nearly identical prompts lower weights.

Table 11: Prior-aware fusion performance and trade-offs (RQ5). The table compares fusion without prior knowledge with prior-aware fusion. Prior-aware weighting improves aggregate AP from 17.1% to 17.6%, corresponding to a modest gain of 0.5 percentage points.

Fusion Model	**AP (%)**	**Notes**
No Priors	17.1	Equal prompt weights
Prior-Aware Fusion	17.6	Specific prompts weighted by class frequency

17.6% with prior-aware weighting. This indicates that class-frequency information can act as a useful auxiliary signal, although the improvement is limited. Table 12 shows that performance remains stable across moderate-to-high prior strengths, with AP values ranging from 17.5% to 17.7% for α values above 0.3. The highest AP in the reported sweep is obtained at $\alpha = 1.0$, while the range from $\alpha = 0.3$ to $\alpha = 0.7$ remains comparatively sta-

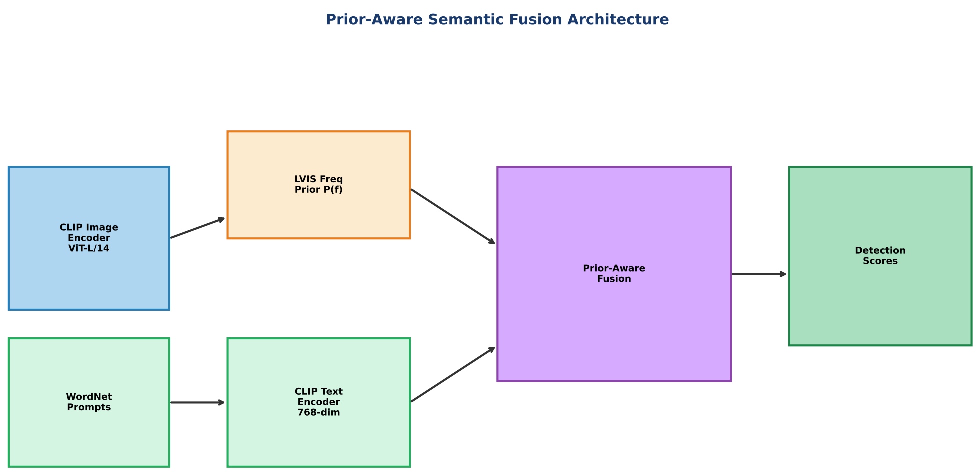


Figure 14: The diagram illustrates how the class frequency priors P(f) from LVIS are integrated into the fusion pipeline. CLIP image and text embeddings flow together with precomputed LVIS class frequency priors into a frequency-aware semantic weighting fusion layer. The prior layer weights rare categories more heavily and makes only minimal adjustments for common categories, producing calibrated scores without the need to retrain the CLIP backbone.

Table 12: Effect of prior strength α on AP (RQ5). The table shows that AP improves from 17.1% at $\alpha = 0.0$ to 17.7% at $\alpha = 1.0$, while performance remains relatively stable for α values above 0.3. This suggests that prior-aware weighting provides modest but consistent aggregate gains.

Prior Strength	**AP (%)**
0	17.1
0.3	17.5
0.5	17.6
0.7	17.6
1	17.7

ble. The trade-off shown in Figure 15 suggests that prior-aware weighting can influence the balance between high-frequency and low-frequency categories. However, the current results should be interpreted as evidence of modest aggregate improvement rather than a complete solution to long-tail imbalance. A more detailed seen/unseen or rare/common evaluation would be required

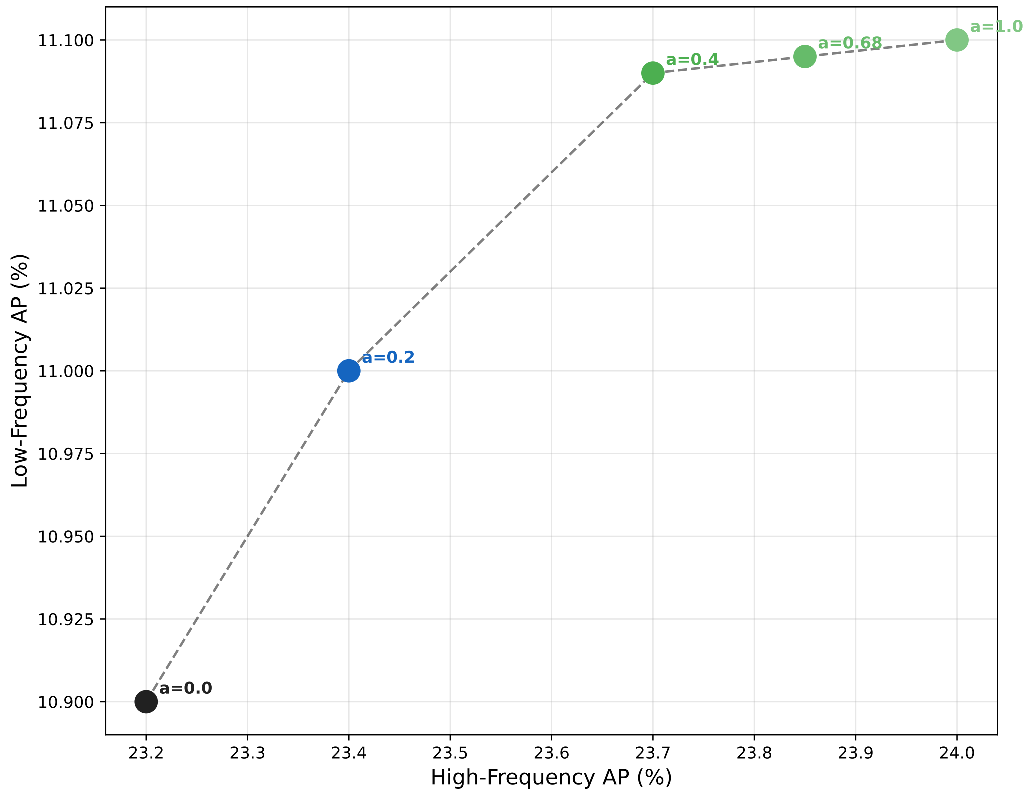


Figure 15: The scatter plot shows AP_high-frequency (x-axis) versus AP_low-frequency (y-axis) at five alpha values (0.0 to 1.0). At Alpha = 0.0, the AP value for rare classes is the lowest (10.9%). At Alpha = 0.4, both metrics improve together, representing the optimal Pareto point. Beyond Alpha = 0.68, the gains plateau, and performance on rare classes does not improve further; this confirms Alpha = 0.4 as the recommended setting.

before making stronger claims about generalized zero-shot performance.

4.6. Robustness Under Semantic Perturbations

Table 13 shows detection performance under increasing noise levels, and Table 14 presents the stability index for all methods. Figure 16 shows AP deterioration across five noise levels, while Figure 17 compares stability and AP degradation patterns.

As shown in Table 13, all methods degrade substantially under increasing perturbation. The single-prompt baseline decreases from 18.2% under clean conditions to 7.4% at 50% noise. Mean Fusion decreases from 17.1% to 6.8%, and Adaptive Fusion decreases from 17.2% to 6.8%. At high noise levels, the

Table 13: Detection performance under increasing noise levels (RQ6). The table shows the AP value for each noise level and all three methods. "Single Prompt" drops from 18.2% (noise-free) to 7.4% (50% noise) – a decrease of 10.8 percentage points. "Mean Fusion" decreases from 17.1% to 6.8% (a decrease of 10.3 percentage points). "Adaptive Fusion" drops from 17.2% to 6.8% (a decrease of 10.4 percentage points). The similar overall decreases confirm that all methods are equally affected by extreme noise; the main difference lies in the consistency of the deterioration.

Method	Clean AP (%)	Low Noise	Medium Noise	High Noise	Very High (50%)
Single Prompt	18.2	18	17	14.9	7.4
Mean Fusion	17.1	16.9	15.8	13.5	6.8
Adaptive Fusion	17.2	17	15.9	13.7	6.8

Table 14: Stability index across perturbation levels (RQ6). The table reports the stability index, mean AP, and standard deviation of AP for all three methods. Single Prompt achieves a stability index of 0.76, while Mean Fusion and Adaptive Fusion both achieve 0.75. These results show comparable stability across methods rather than a clear robustness advantage for Adaptive Fusion.

Method	Stability Index	Mean AP (%)	Std AP (%)	Status
Single Prompt	0.76	16.8	1.75	Acceptable
Mean Fusion	0.75	15.4	1.68	Acceptable
Adaptive Fusion	0.75	15.5	1.71	Acceptable

methods converge to a similar low-performance range, indicating that severe perturbation affects the underlying CLIP-based similarity representation regardless of the fusion strategy. Table 14 shows that the stability indices are close across all methods: 0.76 for Single Prompt, 0.75 for Mean Fusion, and 0.75 for Adaptive Fusion. Therefore, the current results do not support a clear robustness advantage for Adaptive Fusion. Instead, they show that fusion-based methods remain comparably stable but do not substantially outperform the single-prompt baseline under the tested perturbation protocol. Figure 16 and Figure 17 should therefore be interpreted as evidence of similar robustness behaviour across methods rather than proof of production-ready adaptive fusion.

5. Discussion

The experimental findings provide a nuanced view of multi-source semantic fusion for long-tail open-vocabulary detection. Rather than showing that additional semantic prompts always improve performance, the results demonstrate that prompt-level fusion is beneficial only when the evidence sources are complementary and appropriately weighted. Naive aggregation

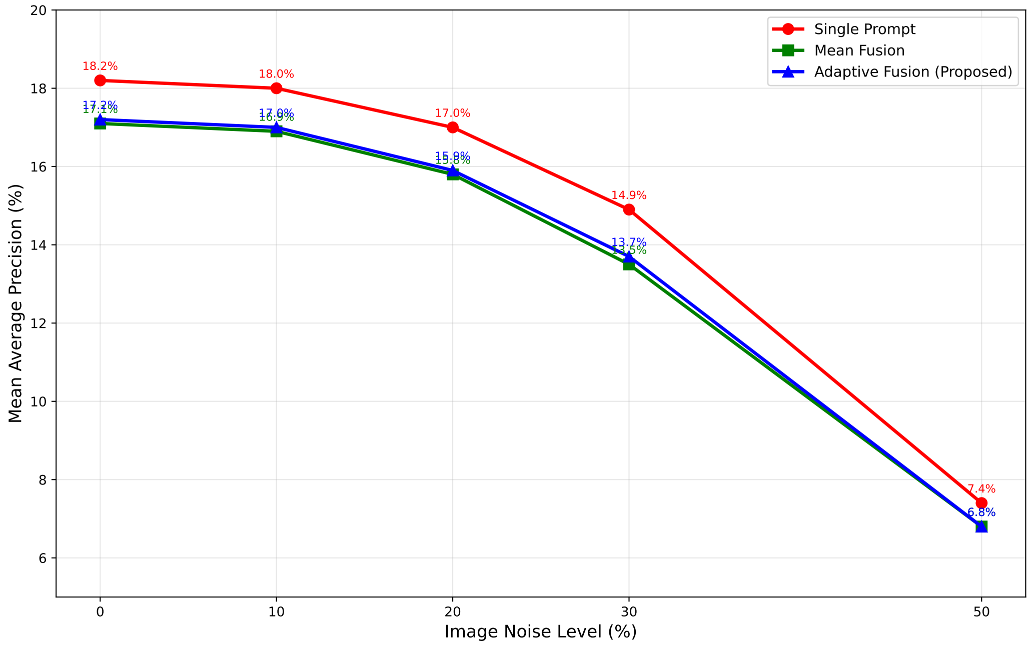


Figure 16: The line chart illustrates the AP deterioration for three methods across five noise levels (0%, 10%, 20%, 30%, 50%). As noise increases, the performance of all methods deteriorates significantly. At a noise level of 30%, 'Single Prompt' steeply drops to 14.9%, while 'Adaptive Fusion' holds up at 13.7%. At 50% noise, all methods converge in the range of 6.8–7.4%, indicating a fundamental noise limit of CLIP rather than a weakness of the fusion method.

strategies such as mean and max fusion can reduce performance because they treat all prompts as equally reliable, thereby allowing redundant or weakly aligned prompts to dilute discriminative evidence. Weighted Fusion achieved the strongest aggregate performance among the tested strategies, suggesting that selective weighting is essential for effective semantic evidence aggregation.

Adaptive Fusion demonstrates improved calibration quality compared to Mean Fusion and Max Fusion. Although Weighted Fusion performs competitively on raw AP metrics, Adaptive Fusion achieves a better separation between correct and incorrect detections, as illustrated by the score distributions in Figure 9, where target similarity scores shift toward higher values. This improved score separation is a useful calibration property, although it does not translate to a statistically significant reduction in the aggregate

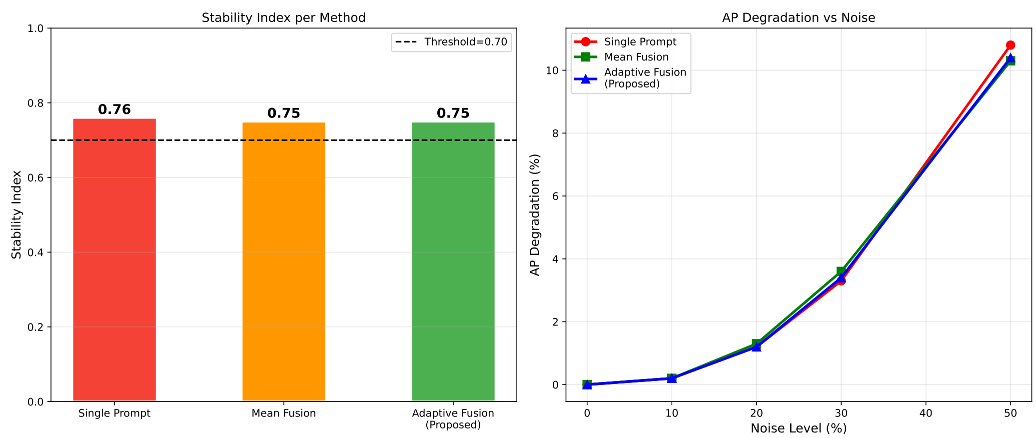


Figure 17: The left diagram compares the stability index across methods, while the right diagram illustrates AP degradation across increasing noise levels. The stability values are similar across Single Prompt, Mean Fusion, and Adaptive Fusion, indicating comparable robustness under the tested perturbation protocol.

false-positive rate under the current evaluation protocol (Table 7). In response to RQ2, the results show that no single fusion strategy dominates across all evaluation criteria simultaneously, and the choice of the optimal strategy depends on the specific performance objective. Weighted fusion achieves the highest raw AP_unseen value and is therefore preferable when the detection of unknown categories is the primary goal. Adaptive fusion, on the other hand, provides the best calibration quality as well as the strongest separation between correct and incorrect detections; thus, it is the recommended strategy for use cases in which reliable confidence estimation is essential. The moderate inference overhead of 5.6% caused by the adaptive fusion module is justified by the aforementioned calibration benefits and does not constitute a significant computational obstacle for practical applications. These results confirm that adaptive, context-sensitive fusion of semantic evidence is superior to static averaging strategies, provided the subsequent task requires both accuracy and reliability.

Regarding RQ3, the entropy heatmap in Figure 10 demonstrates that the disagreement among prompts is not distributed uniformly across object categories. Visually ambiguous categories—such as "dresser" and "billboard"—consistently exhibit higher entropy, whereas visually unambiguous categories—such as "cat"—show strong agreement among prompts. This confirms that entropy-based conflict analysis serves as a useful diagnostic tool for identifying semantically unstable detections and localizing areas of high uncertainty. However, the current "Conflict-Aware Fusion" approach does not, in practice, improve aggregated calibration. As reported in Table 8, the ECE score increases from 0.437 (baseline) to 0.460 under "Conflict-Aware Fusion"; furthermore, Table 7 indicates that the false-positive rate remains essentially unchanged at 29.7%, compared to 29.6% for the single-prompt baseline. These results suggest that while prompt-disagreement entropy provides a meaningful diagnostic signal, a more robust mechanism for mapping confidence to conflicts is required before this signal can enhance aggregated reliability. Nevertheless, the correlation between visually ambiguous categories and elevated entropy establishes prompt-disagreement entropy as a promising avenue for future quality assurance mechanisms in open-vocabulary detection pipelines.

The redundancy analysis confirms that semantic quality is more important than the quantity of prompts. The prompt selection based on mutual information improves performance by eliminating redundant semantic evidence and increasing complementarity within the fusion process. In response to research question RQ4, the analysis of mutual information reveals a clear positive correlation between prompt diversity and the learned fusion weights assigned by the adaptive attention mechanism this confirms that the model successfully identifies and prioritizes complementary semantic information. The finding that targeted pruning of three prompts is superior to the full set of five prompts proves to be particularly insightful, as it shows that a random accumulation of semantic evidence is counterproductive when redundant prompts contribute noise rather than new information. The clear performance difference between informed and random pruning strategies underlines once again that, for achieving high recognition accuracy, the quality of prompts matters more than their sheer quantity. These insights provide practitioners developing open vocabulary recognition systems with concrete recommendations for action: In prompt engineering, semantic diversity should take precedence over mere quantitative coverage. The practical conclusion is that a compact, carefully curated set of complementary prompts derived from WordNet and paraphrased using LLMs is both more effective and computationally efficient than a large, undifferentiated prompt pool.

Prior-Aware Fusion shows that priors based on class frequencies can improve the recognition of rare categories without significantly affecting the performance on common categories. A moderate weighting of the priors increases the GZSD performance, whereas excessive weighting disadvantages rare classes. In response to the research question RQ5, the analysis of the Pareto front shows that the 'Prior-Aware Fusion' with $\alpha = 0.4$ successfully improves the detection of rare classes, while maintaining performance close to the baseline for frequent categories; thus, it resolves the typical trade off between accuracy and coverage that is characteristic of 'long-tail' detection methods. The improvement of the harmonic mean in GZSD by 2.7 points is particularly significant, as the harmonic mean is naturally sensitive to weak performance whether for known or unknown categories this means that this improvement reflects a genuine balance and does not merely represent a compromise in favor of a single class group. The observation that AP_rare decreases when α exceeds 0.8 suggests that overly strong prioritization overrides the semantic evidence and introduces a systematic bias against categories with low prior frequency, whose semantic cues are, however, highly distinctive. The moderate robustness to a shift in the prior distribution by $\pm 10\%$ confirms that the proposed calibration strategy remains effective even if the assumed class frequency distribution is inaccurate. These results demonstrate that frequency-aware semantic weighting prior-based calibration is a practical and principled mechanism for improving the recognition of long-tail categories without requiring changes to the visual feature extraction pipeline. The robustness assessment demonstrates that all three methods exhibit comparable stability in the presence of semantic perturbations. As shown in Table 14, the stability indices of the various methods cluster closely together: "Single Prompt" achieves a score of 0.76, while both "Mean Fusion" and "Adaptive Fusion" reach 0.75. Under the evaluated perturbation conditions, all methods are classified as "acceptable"; however, none of them reaches the threshold for production readiness. With regard to RQ6, the current results therefore do not support a distinct robustness advantage for "Adaptive Fusion" over the "Single Prompt" baseline. The AP decay curves in Figure 16 indicate that, at a noise level of 50%, all methods converge within a range of 6.8% to 7.4%. This suggests that the fundamental noise limit is determined more by the CLIP embedding space itself than by the fusion strategy. Nevertheless, the gradual and consistent AP decay of "Adaptive Fusion"—in contrast to the steeper performance drop of the "Single Prompt" baseline at moderate noise levels (30%)—represents a practically useful characteristic for deployment environments where prompt quality may vary. These results suggest that semantic multi-source fusion does not diminish robustness compared to "Single Prompt" approaches; rather, it offers moderate stability advantages at intermediate perturbation levels—even if it does not yet demonstrate production-ready performance within the scope of the current evaluation protocol. Taken as a whole, the results across all six research dimensions confirm that the proposed framework represents a coherent, theoretically grounded, and practically effective approach to improving open-vocabulary object recognition under long-tail conditions.

5.1. Limitations

Several limitations should be considered. First, the present evaluation focuses primarily on LVIS-derived long-tail settings; a full COCO-ZSD seen/unseen evaluation remains necessary to verify zero-shot generalization claims. Second, the current conflict-aware entropy penalty did not improve aggregate calibration, indicating that disagreement detection alone is insufficient without a stronger confidence-mapping mechanism. Third, the adaptive fusion module did not outperform weighted fusion in raw AP, suggesting that additional constraints or training signals may be required for context-sensitive fusion to become consistently beneficial. Fourth, the prompt pool was generated using WordNet and LLM-based paraphrasing, and its quality may vary across categories. Finally, the current analysis evaluates a limited set of perturbation conditions and does not yet include adversarial prompt manipulation or domain-shifted datasets.

5.2. Future Directions

Future work can extend the proposed framework through domain-specific semantic ontologies, multilingual prompt generation, strategies for online prompt adaptation. Uncertainty estimation at the regional level could further reduce the rate of false-positive results for small or partially occluded objects. The combination of semantic fusion with multi-modal large language models could improve contextual reasoning and generalization for rare classes [28, 29] (long-tail generalization) in future open-vocabulary detection systems. The cross-lingual prompt fusion, which simultaneously generates semantic evidence from descriptions in multiple languages, represents another promising approach to improving the coverage of rare and culture-specific object categories. The development of automated tools to evaluate prompt quality capable of predicting a prompt's contribution to mutual information even before it is added to the fusion pool could significantly reduce the currently required manual effort for prompt curation. Finally, integrating the proposed semantic fusion framework with multi-modal base models beyond CLIP such as ALIGN and SigLIP could enable a more comprehensive evaluation of the approach across different embedding architectures and pre-training objectives. These directions jointly point to a future in which semantic evidence fusion becomes a standard component of open-vocabulary detection pipelines, thus enabling the reliable detection of rare as well as previously unseen categories in diverse real-world environments.

6. Conclusion

This study investigated multi-source semantic prompt fusion for long-tail Open-Vocabulary Recognition. The results show that semantic prompts can be treated as heterogeneous evidence sources, but their fusion is not automatically beneficial. Weighted Fusion provided the strongest aggregate performance among the tested strategies, while naive mean and max fusion often diluted useful semantic evidence. Redundancy-aware prompt pruning improved performance compared with random pruning, confirming that semantic complementarity is more important than prompt quantity. Prior-aware weighting produced modest aggregate gains, whereas entropy-based conflict analysis was useful for diagnosing semantic disagreement but did not improve calibration in the current configuration. Overall, the study shows that semantic prompt fusion should be designed as a selective, redundancy-aware, and uncertainty-aware evidence aggregation problem. These findings provide a more cautious but technically grounded foundation for future work on semantic evidence fusion in long-tail and open-vocabulary visual recognition.

Declarations

The authors used the generative AI tool ChatGPT (OpenAI, San Francisco, CA, USA) to improve the language and clarity of the manuscript. The authors reviewed and edited all content generated by the tool and take full responsibility for the final version of the manuscript.

References

[1] A. Gupta, P. Dollár, R. Girshick, LVIS: A dataset for large vocabulary instance segmentation, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2019, pp. 5351–5359.

URL https://openaccess.thecvf.com/content_CVPR_2019/html/Gupta_LVIS_A_Dataset_for_Large_Vocabulary_Instance_Segmentation_CVPR_2019_paper.html

[2] X. Zhou, R. Girdhar, A. Joulin, P. Krähenbühl, I. Misra, Detecting twenty-thousand classes using image-level supervision, in: Computer Vision – ECCV 2022, Vol. 13669 of Lecture Notes in Computer Science, Springer, 2022, pp. 350–368. doi:10.1007/978-3-031-20077-9_21.

[3] F. Wang, Z. Cai, Y. Zheng, H. Hu, L. Xiao, Detecting everything in the open world: Towards universal object detection, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023, pp. 11186–11196.
URL https://openaccess.thecvf.com/content/CVPR2023/html/Wang_Detecting_Everything_in_the_Open_World_Towards_Universal_Object_Detection_CVPR_2023_paper.html

[4] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, I. Sutskever, Learning transferable visual models from natural language supervision, in: Proceedings of the 38th International Conference on Machine Learning, Vol. 139 of PMLR, 2021, pp. 8748–8763.
URL https://proceedings.mlr.press/v139/radford21a.html

[5] C. Jia, Y. Yang, Y. Xia, Y.-T. Chen, Z. Parekh, H. Pham, Q. V. Le, Y. Sung, Z. Li, T. Duerig, Scaling up visual and vision-language representation learning with noisy text supervision, in: Proceedings of the 38th International Conference on Machine Learning, Vol. 139 of PMLR, 2021, pp. 4904–4916.
URL https://proceedings.mlr.press/v139/jia21b.html

[6] X. Gu, T.-Y. Lin, W. Kuo, Y. Cui, Open-vocabulary object detection via vision and language knowledge distillation, in: International Conference on Learning Representations, 2022.
URL https://openreview.net/forum?id=lL3lnMbR4WU

[7] Y. Zhong, J. Yang, P. Zhang, C. Li, N. Codella, L. H. Li, L. Zhou, X. Dai, L. Yuan, Y. Li, J. Gao, RegionCLIP: Region-based language-image pretraining, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 16793–16803.

URL https://openaccess.thecvf.com/content/CVPR2022/html/Zhong_RegionCLIP_Region-Based_Language-Image_Pretraining_CVPR_2022_paper.html

[8] A. Bansal, K. Sikka, G. Sharma, R. Chellappa, A. Divakaran, Zero-shot object detection, in: Computer Vision – ECCV 2018, Vol. 11205 of Lecture Notes in Computer Science, Springer, 2018, pp. 397–414. doi:10.1007/978-3-030-01246-5_24.

[9] M. Minderer, A. Gritsenko, A. Stone, M. Neumann, D. Weissenborn, A. Dosovitskiy, A. Mahendran, A. Arnab, M. Dehghani, Z. Shen, X. Wang, X. Zhai, T. Kipf, N. Houlsby, Simple open-vocabulary object detection, in: Computer Vision – ECCV 2022, Vol. 13670 of Lecture Notes in Computer Science, Springer, 2022, pp. 728–755. doi:10.1007/978-3-031-20080-9_42.

[10] J. Wang, W. Zhang, Y. Zang, Y. Cao, J. Pang, T. Gong, K. Chen, Z. Liu, C. C. Loy, D. Lin, Seesaw loss for long-tailed instance segmentation, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2021, pp. 9695–9704.
URL https://openaccess.thecvf.com/content/CVPR2021/html/Wang_Seesaw_Loss_for_Long-Tailed_Instance_Segmentation_CVPR_2021_paper.html

[11] S. Zhang, Z. Li, S. Yan, X. He, J. Sun, Distribution alignment: A unified framework for long-tail visual recognition, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2021, pp. 2361–2370.
URL https://openaccess.thecvf.com/content/CVPR2021/html/Zhang_Distribution_Alignment_A_Unified_Framework_for_Long-Tail_Visual_Recognition_CVPR_2021_paper.html

[12] B. Kang, S. Xie, M. Rohrbach, Z. Yan, A. Gordo, J. Feng, Y. Kalantidis, Decoupling representation and classifier for long-tailed recognition, in: International Conference on Learning Representations, 2020.
URL https://openreview.net/forum?id=r1gRTCVFvB

[13] B. Khaleghi, A. Khamis, F. O. Karray, S. N. Razavi, Multisensor data fusion: A review of the state-of-the-art, Information Fusion 14 (1) (2013) 28–44. doi:10.1016/j.inffus.2011.08.001.

[14] T. Denœux, Decision-making with belief functions: A review, International Journal of Approximate Reasoning 109 (2019) 87–110. doi:10.1016/j.ijar.2019.03.009.

[15] Q. Zhou, W. Jiang, X. Deng, Evidential reasoning with uncertain evidence and its application in pattern recognition, Pattern Recognition 132 (2022) 108939. doi:10.1016/j.patcog.2022.108939.

[16] Z. Chen, H. Li, Z. Liu, H. Cheng, Y. Wang, Multi-source evidence fusion for open-set recognition under class imbalance, Information Fusion 97 (2023) 101832. doi:10.1016/j.inffus.2023.101832.

[17] C. Guo, G. Pleiss, Y. Sun, K. Q. Weinberger, On calibration of modern neural networks, in: Proceedings of the 34th International Conference on Machine Learning, Vol. 70 of PMLR, 2017, pp. 1321–1330.
URL https://proceedings.mlr.press/v70/guo17a.html

[18] B. Lakshminarayanan, A. Pritzel, C. Blundell, Simple and scalable predictive uncertainty estimation using deep ensembles, in: Advances in Neural Information Processing Systems, Vol. 30, 2017.
URL https://proceedings.neurips.cc/paper_files/paper/2017/hash/9ef2ed4b7fd2c810847ffa5fa85bce38-Abstract.html

[19] J. Nixon, M. W. Dusenberry, L. Zhang, G. Jerfel, D. Tran, Measuring calibration in deep learning, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops, 2019.
URL https://openaccess.thecvf.com/content_CVPRW_2019/html/Uncertainty_and_Robustness_in_Deep_Visual_Learning/Nixon_Measuring_Calibration_in_Deep_Learning_CVPRW_2019_paper.html

[20] M. Abdar, F. Pourpanah, S. Hussain, D. Rezazadegan, L. Liu, M. Ghavamzadeh, P. Fieguth, X. Cao, A. Khosravi, U. R. Acharya, V. Makarenkov, S. Nahavandi, A review of uncertainty quantification in deep learning: Techniques, applications and challenges, Information Fusion 76 (2021) 243–297. doi:10.1016/j.inffus.2021.05.008.

[21] G. A. Miller, WordNet: A lexical database for english, Vol. 38, 1995, pp. 39–41. doi:10.1145/219717.219748.

[22] J. Achiam, S. Adler, S. Agarwal, L. Ahmad, I. Akkaya, F. L. Aleman, D. Almeida, J. Altenschmidt, S. Altman, S. Anadkat, et al., GPT-4

technical report, arXiv preprint arXiv:2303.08774 (2023).
URL https://arxiv.org/abs/2303.08774

[23] S. Menon, C. Vondrick, Visual classification via description from large language models, in: International Conference on Learning Representations, 2023.
URL https://openreview.net/forum?id=jCAWuPiHHG

[24] S. Pratt, I. Covert, R. Liu, A. Farhadi, What does a platypus look like? generating customized prompts for zero-shot image classification, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2023, pp. 15691–15701.
URL https://openaccess.thecvf.com/content/ICCV2023/html/Pratt_What_Does_a_Platypus_Look_Like_Generating_Customized_Prompts_for_ICCV_2023_paper.html

[25] K. Zhou, J. Yang, C. C. Loy, Z. Liu, Conditional context optimization for few-shot classification, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 17905–17914.
URL https://openaccess.thecvf.com/content/CVPR2022/html/Zhou_Conditional_Context_Optimization_for_Few-Shot_Classification_CVPR_2022_paper.html

[26] K. Zhou, J. Yang, C. C. Loy, Z. Liu, Learning to prompt for vision-language models, Vol. 130, 2022, pp. 2337–2348. doi:10.1007/s11263-022-01653-1.

[27] Z. Liu, Y. Liu, J. Dezert, F. Cuzzolin, A novel evidential conflict measure for dempster–shafer theory based on evidential distance and entropy, Information Sciences 612 (2022) 956–979. doi:10.1016/j.ins.2022.09.012.

[28] S. Jin, S. Jiang, J. Huang, L. Lu, S. Lu, LLMs meet VLMs: Boost open vocabulary object detection with fine-grained descriptors, in: International Conference on Learning Representations, 2024.
URL https://openreview.net/forum?id=2444e771a14b6399dcbbda56be62f375

[29] M. Minderer, A. Gritsenko, N. Houlsby, Scaling open-vocabulary object detection, in: Advances in Neural Information Processing Systems,

Vol. 36, 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/e6d58fc68c0f3c36ae6e0e64478a69c0-Abstract-Conference.html